

SCHOOL OF ENGINEERING, CUSAT
B. TECH DEGREE 2ND SEMESTER SECOND INTERNAL EXAMINATIONS, JULY 2024
23-200-0204B OBJECT ORIENTED PROGRAMMING IN C++
(CS A/B, EE-B)
(2023 Scheme)

TIME: 2 hrs

MAX. MARKS: 30

Course Outcomes

On completion of this course the student will be able to:

CO1	Understand the basic concepts of object oriented programming
CO2	Describe the object oriented programming with concepts of streams, classes, functions, data and objects
CO3	Implement object oriented programming construct like encapsulation, inheritance and polymorphism.
CO4	Understand dynamic memory management technique using pointers, constructors, destructors
CO5	Identify classes including data, methods and the relationship among the classes from a given statement and solve problem using object oriented constructs in C++.

PART A

(Answer all questions)

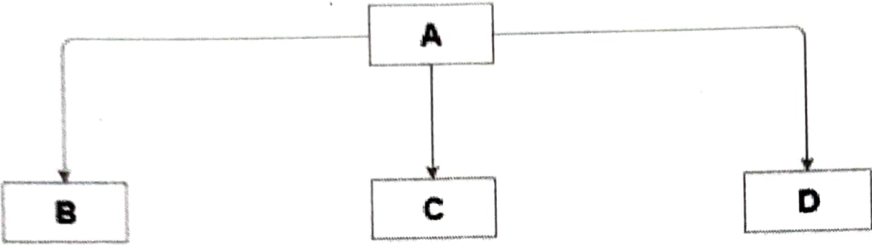
(5 x 2= 10 Marks)

	Questions	Marks	BL	CO	PO
I (a)	Explain Virtual function with example.	2	L1	3	1,2,3
(b)	List the operator that can't be overloaded.	2	L2	4	3,12
(c)	How polymorphism is achieved at a) Compile time b)Run time	2	L1	2	1,2
(d)	Write a short note on file pointers and their manipulation functions.	2	L1	2	1,2
(e)	Differentiate between the properties of the following two derived classes. Class D1:private B Class D2:public B	2	L1	3	3,1,2

PART B

(Answer the following questions
choosing either the first OR second option)

(1 x 10= 10 Marks)

Q.No.	Questions	Marks	BL	CO	PO
II	Identify the inheritance and write an example program. 	10	L1	3	1,2,3
OR					
III	Unary operators overloaded by means of a member function take no explicit arguments and return no explicit values. Justify your answer with program.	10	L1	3	1,2,3
(1 x 10= 10 Marks)					
IV	We have two classes Time and Minute. If 'm' is an object of Minute Class and 't' is an object of Time Class and we want convert Time Class to Minute Class. Produce the conversion routine with example program.	10	L3	5	3,12
OR					
V	Explain under what circumstance the following statement would be used (write the programs.). a)throw b)Catch(...)	10	L3	5	3,12